Quiz 5

Graded

Student

Myles Sartor

Total Points

13 / 14 pts

Question 1

(no title) **6** / 6 pts

✔ **– 0 pts** Correct

– 0.5 pts algebra error

– 0.5 pts part a: final answer needs to be in at least Φ notation if not a final number

– 1 pt part a: Φ used incorrectly

– 0.5 pts part b: final answer needs to be in at least Φ notation if not a final number

– 1 pt part a: misunderstood what probability was being asked for

– 3 pts part a: missing

– 2 pts Did not find the correct distribution for $\bar{X}$

– 1 pt part b: Φ used incorrectly

– 1 pt part b: misunderstood what probability was being asked for

– 3 pts part b missing

– 1.5 pts part b: incorrect and no work shown

– 1 pt part a: error in doing work

– 1 pt part b: error in doing work

Question 2

Problem 2

7 / 8 pts

- **– 0 pts** Correct
- **– 1 pt** Expect value $\mathbb{E}[x] = 1.5$: incorrect setup
- **– 1 pt** Expect value $\mathbb{E}[x] = 1.5$: incorrect answer
- **– 1 pt** Expect value $\mathbb{E}[x^2] = 3$
- **– 1 pt** $Var(X) = \mathbb{E}[x^2] - \mathbb{E}[x]^2 = \frac{3}{4}$: incorrect setup
- **– 1 pt** $Var(X) = \mathbb{E}[x^2] - \mathbb{E}[x]^2 = \frac{3}{4}$: incorrect answer
- **– 1 pt** sample variance is $\frac{3}{4*300} = \frac{1}{400}$
- **– 0.5 pts** z score for 1.6 is $\frac{1.6-1.5}{\sqrt{\frac{1}{400}}} = 2$
- **– 0.5 pts** z score for 1.65 is $\frac{1.65-1.5}{\sqrt{\frac{1}{400}}} = 3$
- ✔ **– 1 pt** $P(1.6 < \bar{X}_n < 1.65) = P(2 < z < 3) = \phi(3) - \phi(2) = (0.997 - 0.95)/2 = 0.0235$
- **– 1 pt** sample expected value is $\mathbb{E}[x]$
- **– 0.5 pts** algebraic error
- **– 8 pts** incorrect
- **– 0.5 pts** Used 95 and 99.7 instead of 0.95 and 0.997

Quiz 5

Name: Myles Sartor Student ID: 119017708

Problem 1. The university dining team wants to know how much students spend on campus meals in a semester. Suppose the population mean is \$500 and the population standard deviation is \$20. A random sample of 1600 students is chosen.

(a) What is the probability that the sample mean will be within \$1 of \$500?

(b) Find the probability that the sample mean will exceed \$550?

$\sigma = 20$

$\mu = 500$

$n = 1600$

$$\frac{\bar{X}-\mu}{\sigma^2/n} \rightarrow \frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$$

a) $P(500-1 < \bar{X} < 500+1)$

$= P(499 < \bar{X} < 501)$

$= P\left(\frac{499-500}{20/\sqrt{1600}} < Z < \frac{501-500}{20/\sqrt{1600}}\right)$

$= P\left(\frac{-1}{20/40} < Z < \frac{1}{20/40}\right)$

$= P(-2 < Z < 2)$

$= \Phi(2) - \Phi(-2) = \boxed{0.95}$ by empirical rule

b) $P(\bar{X} > 550) = 1 - P\left(Z < \frac{550-500}{20/40}\right)$

$$\frac{\frac{50}{1}}{\frac{20}{40}} = \frac{50}{1} \cdot \frac{40}{20} = 100$$

$1 - P(Z < 100)$

$= 1 - \Phi(100)$

Quiz 5

Name: Myles Sartel — Student ID: 119017708

Problem 2. Let $X_1, X_2, \cdots, X_{300}$ be a random sample of size $n = 300$ from a distribution with probability density function

$$f(x) = \begin{cases} \dfrac{3}{x^4} & x > 1 \\ 0 & \text{otherwise} \end{cases}$$

Calculate $P(1.6 < \bar{X}_n < 1.65)$ using central limit theorem.
Hint: Begin by calculating μ and σ^2.
Hint: If needed at the end use 68-95-99.7 Rule.

L'Hopital:

$3x^{-3} \to \frac{3x^{-2}}{-2}$ $\quad -\frac{3}{2x^2}$

$$E[x] = \int_1^\infty x \cdot \frac{3}{x^4}\,dx = \int_1^\infty \frac{3x}{x^4}\,dx = \int_1^\infty \frac{3}{x^3}\,dx = -\frac{3}{2}x^{-2}\Big|_1^\infty$$

$$= \frac{3}{2} = E[\bar{x}]$$

$3x^{-2} = \frac{3x^{-1}}{-1} = -\frac{3}{x}$

$$E[x^2] = \int_1^\infty x^2 \cdot \frac{3}{x^4}\,dx = \int_1^\infty \frac{3x^2}{x^4}\,dx = \int_1^\infty \frac{3}{x^2}\,dx = -\frac{3}{x}\Big|_1^\infty$$

$$= 3 = E[\bar{x}^2]$$

$$\sigma^2 = var(x) = E[x^2] - (E[x])^2 = 3 - \left(\frac{3}{2}\right)^2 = 3 - \frac{9}{4} = \frac{12}{4} - \frac{9}{4} = \frac{3}{4}$$

$$X \sim N\left(\mu = \frac{3}{2}, \sigma^2 = \frac{3}{4}\right)$$

$$\frac{\frac{3}{4}}{\frac{300}{1}} = \frac{3}{4} \cdot \frac{1}{300} = \frac{3}{1200} = \frac{1}{400}$$

$$P(1.6 < \bar{x}_n < 1.65) =$$

$$\frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$

$$P\left(\frac{1.6 - 1.5}{\sqrt{\frac{3/4}{300}}} < Z < \frac{1.65 - 1.5}{\sqrt{\frac{3/4}{300}}}\right) = P\left(\frac{0.1}{\sqrt{1/400}} < Z < \frac{0.15}{\sqrt{1/400}}\right)$$

$$\frac{\frac{1}{10}}{\frac{1}{20}} = \frac{1}{10} \cdot \frac{20}{1} = 2$$

$$= P\left(\frac{0.1}{1/20} < Z < \frac{0.15}{1/20}\right)$$

$$\frac{\frac{15}{100}}{\frac{1}{20}} = \frac{15}{100} \cdot \frac{20}{1} = \frac{300}{100} = 3$$

$$= P(2 < Z < 3) = 0.99.7 - 0.95 = \boxed{0.047}$$

$$\phi(3) - \phi(2)$$